PROJECT 3 -
Salient Object Detection

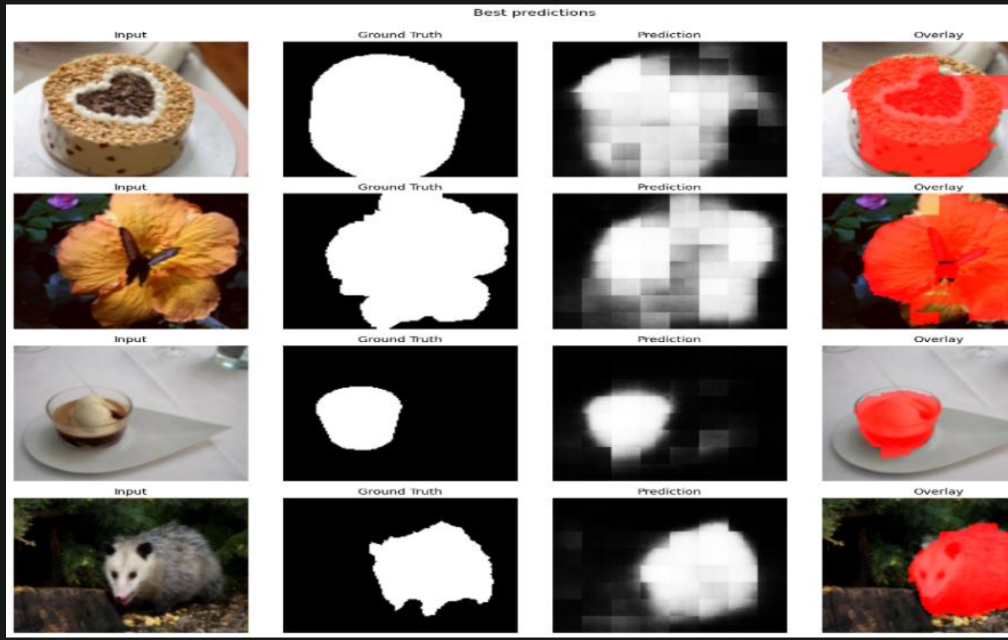
Dataset: DUTS Dataset

- 10,553 images and masks
- 70% Train / 15% Validation / 15% Test
- Image size: 224x224
- Data augmentation:
    - Horizontal flip
    - Brightness adjustment
    - Random cropping

```
Split -> Train: 7387 | Val: 1582 | Test: 1584
Image batch: torch.Size([8, 3, 224, 224]) range: 0.0 1.0
Mask batch : torch.Size([8, 1, 224, 224]) range: 0.0 1.0
```

Model Architecture

- CNN Encoder-Decoder architecture
- Convolutional layers + ReLU + MaxPooling
- Transposed Convolutions for decoding
- Batch Normalization and Dropout



Training

- Loss Function: $BCE + 0.5 \times (1 - IoU)$
- Optimizer: Adam• Epochs: 25
- Trained using Google Colab GPU

Final Results

- IoU: 70%
- Precision: 78%
- Recall: 86%
- F1 Score: 79%

The model achieved strong segmentation performance on the DUTS test dataset.

```
Loaded best checkpoint from epoch 22.
Evaluating on test set: 100%  198/198 [00:18<00:00, 11.69it/s]

=========== Test Set Results ===========
Samples evaluated : 1584
IoU        : 0.7098
Precision  : 0.7757
Recall     : 0.8593
F1 Score   : 0.7872
========================================
```

Conclusion

- Successfully implemented a Salient Object Detection system
- Built and trained completely from scratch
- Demonstrated practical knowledge in:
  - Computer Vision
  - Deep Learning
  - Image Segmentation

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

✅ Model loaded.
```